SplitStack Build and Deployment Instructions

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Prerequisites

- Node.js 22 LTS.
- npm, included with Node.js.
- Flutter SDK for the mobile app.
- Chrome for the quickest Flutter web/mobile demo.
- Xcode for iOS Simulator.
- Android Studio and Android SDK for Android Emulator.

Backend Setup

From the project root:

```
cd website/backend
npm install
npm run dev
```

Expected backend URL:

```
http://localhost:3001
```

Health check:

```
curl http://localhost:3001/api/health
```

Expected health response includes:

```
{
  "ok": true,
  "service": "SplitStack API",
  "database": "sqlite"
}
```

Backend Environment Variables

Create `website/backend/.env` when AI assistant calls should use Gemini:

```
GEMINI_API_KEY=your-gemini-api-key
GEMINI_MODEL=gemini-2.5-flash
PORT=3001
```

If `GEMINI_API_KEY` is omitted, the backend returns local SplitStack assistant summaries.

Web Frontend Setup

Open a second terminal:

```
cd website/frontend
npm install
npm run dev
```

Expected frontend URL:

```
http://localhost:5173
```

The frontend expects the backend to be running at `http://localhost:3001`.

Production Build Checks

Backend:

```
cd website/backend
npm run lint
```

Frontend:

```
cd website/frontend
npm run lint
npm run build
```

On Windows PowerShell, if script execution blocks `npm`, use:

```
npm.cmd run lint
npm.cmd run build
```

Mobile App Setup

Start the backend first, then open another terminal:

```
cd mobile-app
flutter pub get
flutter run
```

For the quickest local demo, choose Chrome when Flutter asks for a device:

```
flutter run -d chrome
```

Other common targets:

```
flutter run -d ios
flutter run -d android
flutter run -d macos
```

Mobile API Base URLs

Target	Default API URL
Chrome	`http://localhost:3001`
iOS Simulator	`http://127.0.0.1:3001`
Android Emulator	`http://10.0.2.2:3001`
macOS desktop	`http://127.0.0.1:3001`

For a physical phone, put the phone and laptop on the same Wi-Fi, find the laptop IP, then pass it to Flutter:

```
flutter run -d <device-id> --dart-define=API_BASE_URL=http://YOUR_IP:3001
```

Demo Deployment Flow

1. Start the backend with `npm run dev`.
2. Start the web frontend with `npm run dev`.
3. Optionally start the mobile app with `flutter run -d chrome`.
4. Register a new account.
5. Create a group and invite another account by email.
6. Log in as the invited account and accept the invite.
7. Add normal and above-threshold expenses.
8. Approve or decline the generated vote.
9. Record a settlement payment.
10. Create a challenge and add contribution progress.
11. Upload a receipt image.

.2. Set a monthly budget and review analytics.
.3. Ask the assistant about balances, spending, or budget recommendations.

Troubleshooting

Issue	Resolution
`Cannot reach backend`	Confirm the backend is running on port 3001.
Frontend opens but API calls fail	Check the backend terminal and browser console.
Port already in use	Run the backend with `PORT=3002 npm run dev` and update client configuration if needed.
PowerShell blocks npm	Use `npm.cmd` commands or adjust execution policy for your shell.
Flutter cannot see backend on Android Emulator	Use `http://10.0.2.2:3001` or pass `API_BASE_URL`.
Gemini assistant does not call API	Add `GEMINI_API_KEY` to `website/backend/.env` and restart backend.